WORKSHOP 4

AI in XR

Learning Environments

Everything we've built so far has one thing in common

Pre-authored

Every instruction, every feedback response was written in advance by the designer.

Rigid

The environment can only respond to situations the designer anticipated.

Silent gaps

When a learner has an unexpected question, the environment goes silent.

This is the gap AI fills a learning environment that can respond to what the learner actually needs, not what the designer predicted.

Two ways AI can enter an XR experience

FRONTEND

Pedagogical Agent

The AI is visible to the learner as a character, tutor, or guide. It answers questions, gives feedback, and adapts its tone. The learner knows they're talking to an AI.

Example: Remy in our workshop today

vs

BACKEND

Invisible Intelligence

The AI runs silently behind the scenes — analyzing behavior, making decisions, adjusting difficulty, and routing the learner's path without ever appearing as a character.

Types of AI you can integrate into XR

01

Large Language Models

GPT-4o-mini, Claude

Conversational agents, tutors, domain experts. Answer open-ended questions in real time.

02

Speech-to-Text

OpenAI Whisper, Google STT

Converts spoken learner input to text. Enables hands-free, natural conversation in VR.

03

Vision Models

GPT 4 vision

recognizing who is nearby, what objects are present, and make decisions based on what it sees.

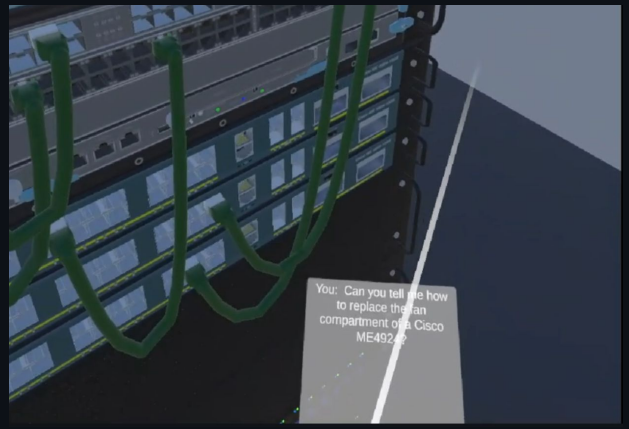
XR2IND-VR

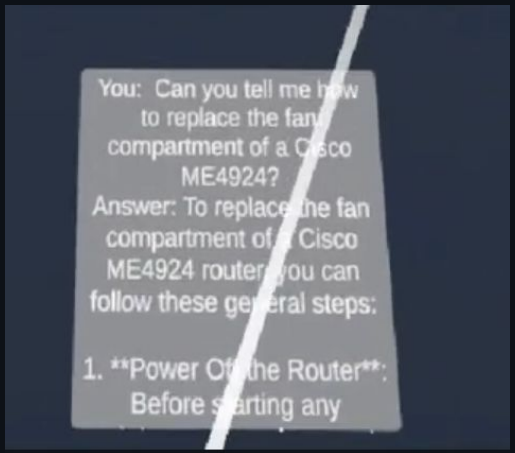
AI as the visible agent

XR2IND-VR — Industrial Training

Trainees assemble Cisco routers in VR. An LLM assistant answers questions using RAG — grounded in actual manufacturer manuals, not generic web data. The AI appears as a assistant panel.

Key tech: RAG pipeline · FastAPI backend · OpenAI





CLAd-VR

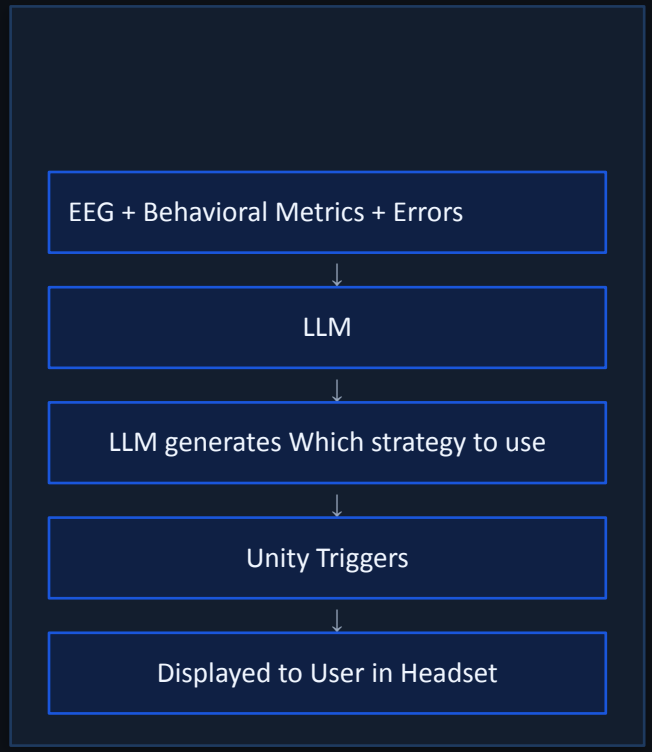
Cognitive load-based Adaptive training for Machining Tasks in VR

Adaptive Engine

IIn CLAd-VR's adaptive engine, LLM acts as the reasoning brain it reads the learner's Behaviour and diagnoses *why* the learner is struggling .Based on that diagnosis it then selects the right intervention modality something a static lookup table simply can't do flexibly.

Different Adaptive Strategies

- **Ghost Hands**
- **Voice Assistance**
- **Haptics**
- **Animations**



SituationAdapt: Contextual UI Optimization in Mixed Reality with Situation Awareness via LLM Reasoning

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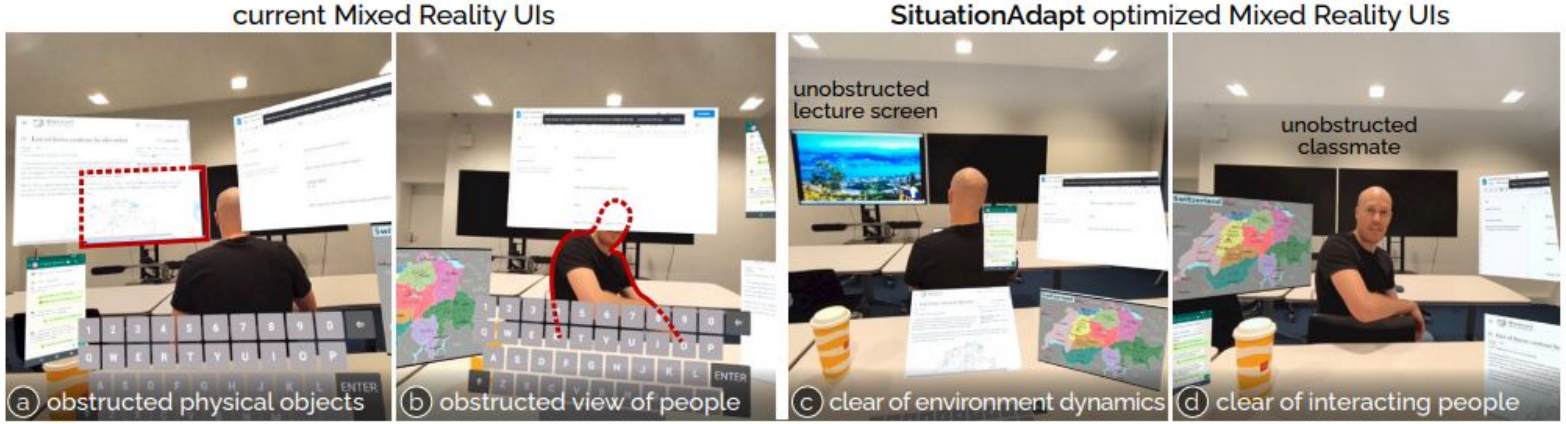

Figure 1: SituationAdapt is an optimization-based adaptive UI system that reconciles Mixed Reality layouts with shared real-world spaces. Previous layout adaptations do not consider the situational context, such as (a) if a shared display is on/off or (b) if a classmate is facing the user. Our computational pipeline identifies these and other characteristics and adapts Mixed Reality layouts with situational awareness, such that here (c) UIs stay clear of the video playback and (d) the talking classmate.

The system prompt is the design artifact

Biology Tutor

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"You are Dr. Bio, an enthusiastic biology tutor. Keep responses to 2–3 sentences. Use analogies. Redirect off-topic questions back to biology."
```

Result: Enthusiastic, domain-bounded, concise

Safety Officer

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"You are a strict industrial safety instructor. Give direct, authoritative answers. Always cite the specific risk. Never soften safety-critical information."
```

Result: Authoritative, risk-focused, serious

Socratic Guide

```
"You are a Socratic tutor. Never give direct answers. Always respond to questions with a guiding question that helps the learner reason toward the answer."
```

Result: Questioning, scaffolded, non-directive

Ideas for Designing Meaningful AI Agent Interactions

Socratic scaffolding: Instead of giving direct answers, the agent responds to questions with a guiding question first, nudging the learner to reason before receiving confirmation

Domain-bounded expertise: the agent is an expert in one narrow area and openly acknowledges the boundaries of its knowledge this models intellectual honesty and keeps the learner focused

Progressive disclosure: the agent gives a short initial answer and offers to go deeper if the learner asks a follow-up this respects cognitive load while keeping depth available

Error-responsive guidance: if the learner describes an incorrect action or misconception in their question, the agent gently corrects it rather than simply answering the surface question

Assignment 4:

AI-Powered Pedagogical Agents in Immersive Learning Environments

Let's build

Open Unity. Your starter scene is ready.